

FinalProject_group010-Fa22

May 29, 2026

1 COGS 108 - Analyzing The Relationship between Mean Income & EV Population

Video Link: https://drive.google.com/file/d/1uJoChhRjFG6_VSAN7j0w4m4eFZ8UjYlt/view?usp=share_link

1.1 Permissions

Place an X in the appropriate bracket below to specify if you would like your group's project to be made available to the public. (Note that student names will be included (but PIDs will be scraped from any groups who include their PIDs)).

- ☒ YES - make available
- ☐ NO - keep private

2 Names

- Bo Li PID: A17239888
- Shan He PID: A17239809
- Michael Schaar PID: A16620164
- Saman Zarei PID: A16613221

3 Overview

- This project focuses on the relationship between the registration of electric and hybrid vehicles and house income in the states of New York and Texas. We want to determine if there is a relationship between house income and the number of electric and hybrid vehicles being registered. If a higher income is associated with more vehicle registrations we hope to find a positive correlation between income and vehicle registration population, and vice-versa that a lower income is associated with a lower population of registered electric and hybrid vehicles. We also want to see the differences between these two states through the use of other interesting observations.

Research Question

- How does the mean house income by zip code correlate with the population of Electric and Hybrid vehicles in the States of New York State and Texas, and are there any differences between these two states.

4 Background & Prior Work

- The invention of electric vehicles has always been controversial in terms of clean energy considerations. Many people who agree with the idea are more likely to show their willingness to own EVs. In recent years, more and more EV companies appeared in the vehicle market. Popular ones include Tesla, with a sales number of 172,700; Ford Mustang March-E, with a sales number of 27,140; Chevrolet Bolt, with a sales number of 24,828; and Volkswagen, etc.; some areas in the US have been populated by more EVs than other parts of the US such as California. However, since the majority of people already own gas cars so, changing to EVs would not be a money-wise decision. That means such a decision involves financial consideration.
- In one of the previous research studies, a report from CarGurus shows that 53% of active car shoppers also are considering buying electric cars. The data that was collected by Time shows the expectation of buying electric cars in the next 5 years from 2018 was 15%, and in 2022 this amount has reached 40%. Also, the plan that people had to buy electric cars in the next 10 years was 34%. More and more people expect to buy electric cars in the next few years. One study research provided by Time is that increasing Tesla has made profits of about \$3.32 billion only in the first 3 months of 2022; the percentage of people who are willing to buy electric cars is going up, so this amount till 2021 has reached 56%. With all the information we studied now, we want to dig deeper and gather more data to analyze if there is a relationship between house income and the sales number of electric and hybrid vehicles within and between different states. If so, how does one affect another?

References:

- <https://www.kbb.com/best-cars/most-popular-electric-cars/>
- <https://www.cnn.com/2022/08/05/business/ev-hybrid-shopping-gas-prices/index.html>

5 Hypothesis

- HYPOTHESIS: There is a correlation between the number of Electric and Hybrid vehicles and the house income in New York State and Texas.
- NULL HYPOTHESIS: There is no difference between the correlation of the number of Electric and Hybrid vehicles and the mean house income in New York State and Texas.
- ALTERNATIVE HYPOTHESIS: The higher amount of home income influences a higher population of electric vehicles within the State of New York and the State of Texas. The lower amount of home income influences a lower population of electric vehicles within the State of New York and the State of Texas.

6 Dataset(s)

6.0.1 Dataset #1

- Dataset Name: New York State EV Registration up to 08/01/2022
- Link to the dataset: <https://www.atlasevhub.com/materials/state-ev-registration-data/#data>
- Number of observations: 3118179 x 10

Description:

- The dataset above consists of numbers of electric vehicle registration in New York state provided by Atlas, which works directly with states to make data on electric vehicle registration publicly accessible. This data shows the number of registered electric vehicles in New York state sorted by zip code, which shows the model of electric vehicles that people have registered. This dataset shows the popularity of electric vehicles in New York State. Then in, the second part of dataset #1 shows the total number of electric vehicles based on each zip code in New York as “Total_EV_Counts. This amount means the total number of electric vehicles from all models in that zip code. We will use this dataset later to combine it with the income of the people of New York state based on zip code.

6.0.2 Dataset #2

- Dataset Name: Texas EV Registration up to 07/01/2022
- Link to the dataset: <https://www.atlasevhub.com/materials/state-ev-registration-data/#data>
- Number of observations: 3118179 x 10

Description:

- The dataset above consists of numbers of electric vehicle registration in Texas state provided by Atlas, which works directly with states to make data on electric vehicle registration publicly accessible. This data shows the number of registered electric vehicles in Texas state sorted by zip code, which shows the model of electric vehicles that people have registered. This dataset also shows the popularity of electric vehicles in Texas State. We will use this dataset later to combine it with the income of the people of Texas state based on zip code.

6.0.3 Dataset #3

- Dataset Name: Combination of Dataset #1 & #2 with US household static income
- Link to the dataset: <https://www.atlasevhub.com/materials/state-ev-registration-data/#data> https://www.kaggle.com/datasets/goldenoakresearch/us-household-income-stats-geo-locations?select=kaggle_income.csv
- Number of observations: 2105 x 5, 2105 x 5, 2144 x 4, 2144 x 5

Description:

- This data set is the combination of dataset #1, dataset #2, and the US Household Income Static provided by Kaggle, which shows the household income of each state based on zip code. The first part of dataset #3 shows the head of household income in one of the states of the USA, which means this dataset provides us with the household income in each state sorted by each zip code.

Combining datasets

- The second part of dataset #3 is the combination of dataset #1 and US household static income To combine datasets for New York. This dataset shows the mean income of each household (Mean_Income) sorted by zip code and also shows the total number of Electric Vehicle (Total_EV_Counts) in that zip code to see how the income affects the popularity

of electric vehicles in that zip code in New York state. Also, it shows the most popular electric vehicle model in that zip code (Vehicle_Name) and the number of that vehicle model (Vehicle_Counts) in New York state.

- The third part of dataset #3 is the combination of dataset #2 and US household static income To combine datasets for Texas. This dataset shows the mean income of each household (Mean_Income) sorted by zip code and also shows the total number of Electric Vehicle (Total_EV_Counts) in that zip code to see how the income affects the popularity of electric vehicles in that zip code in Texas state. Also, it shows the most popular electric vehicle model in that zip code (Vehicle_Name) and the number of that vehicle model (Vehicle_Counts) in Texas state.

7 Setup

```
[32]: ## YOUR CODE HERE
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import patsy
import statsmodels.api as sm
```

8 Data Cleaning

Describe your data cleaning steps here.

Dataset #1

```
[2]: # import dataset #1
ny_ev = pd.read_csv("ny_ev_registrations_public.csv")
```

C:\Users\Henry\AppData\Local\Temp\ipykernel_20048\3200454009.py:2: DtypeWarning: Columns (5) have mixed types. Specify dtype option on import or set low_memory=False.

```
ny_ev = pd.read_csv("ny_ev_registrations_public.csv")
```

```
[3]: # get an idea of how the dataset looks like
ny_ev.head()
```

```
[3]:   DMV ID      DMV Snapshot  ZIP Code State Registration Valid Date \
0      39  NY DATA.NY.GOV (4/2/2020)    10583    NY           3/27/2019
1      39  NY DATA.NY.GOV (4/2/2020)    10583    NY           12/24/2018
2      39  NY DATA.NY.GOV (4/2/2020)    10583    NY           12/4/2018
3      39  NY DATA.NY.GOV (4/2/2020)    10583    NY           12/1/2018
4      39  NY DATA.NY.GOV (4/2/2020)    10583    NY           11/29/2018
```

```
   Registration Expiration Date VIN Prefix VIN Model Year  Vehicle Name \
0           12/31/2020    5YJ3E1EB          J  Tesla Model 3
```

1	12/23/2020	5YJ3E1EB	J	Tesla Model 3
2	12/3/2020	5YJ3E1EB	J	Tesla Model 3
3	11/30/2020	5YJ3E1EB	J	Tesla Model 3
4	11/28/2020	5YJ3E1EB	J	Tesla Model 3

	Technology
0	BEV
1	BEV
2	BEV
3	BEV
4	BEV

```
[4]: # Keep the columns we need and check if there is any missing data
ny_ev = ny_ev[["State", "ZIP Code", "Vehicle Name"]]
ny_ev.isna().sum() # no null values
```

```
[4]: State          0
ZIP Code          0
Vehicle Name      0
dtype: int64
```

```
[5]: # check the data types in this dataset
ny_ev.dtypes # the data types are good for us to use no modifications needed
```

```
[5]: State          object
ZIP Code          int64
Vehicle Name      object
dtype: object
```

```
[6]: # groupby the dataset ny_ev by Zip code and Vehicle Name
# to get the counts of the vehicles by state
ny_vehicle_count = ny_ev.groupby('ZIP Code')['Vehicle Name']. \
    value_counts().rename('Vehicle_Counts').reset_index()

# create a new dataset counting the total number of evs in each zip code area
total_ev_ny = ny_vehicle_count.groupby('ZIP Code'). \
    sum().reset_index().rename(columns = {"ZIP Code": "Zip_Code", " \
    ↪ Vehicle_Counts": "Total_EV_Counts"})

# create a new dataset showing the most popular model of ev in each zip code_ \
    ↪ area
max_ev_ny = ny_vehicle_count.groupby('ZIP Code').max(). \
    reset_index().rename(columns = {"ZIP Code": "Zip_Code"})
```

Dataset #2

```
[7]: # import dataset #2
tx_ev = pd.read_csv("tx_ev_registrations_public.csv")
```

```
[8]: # get an idea of how the data looks like
tx_ev.head()
```

```
[8]:
```

	Vehicle ID	ZIP Code	Registration	Valid Date	VIN Prefix	DMV_ID	\
0	TX-003-0344972	75060	2021-07-01	00:00:00.000	5YJ3E1EA	3	
1	TX-003-0344978	77005	2021-07-01	00:00:00.000	5YJ3E1EA	3	
2	TX-003-0345022	75039	2021-07-01	00:00:00.000	5YJ3E1EA	3	
3	TX-003-0345034	77006	2021-07-01	00:00:00.000	5YJ3E1EA	3	
4	TX-003-0345040	77094	2021-07-01	00:00:00.000	5YJ3E1EA	3	

	DMV ID Complete	DMV Snapshot	\
0	TX3	TX Registration Data from DFW Clean Cities (20...	
1	TX3	TX Registration Data from DFW Clean Cities (20...	
2	TX3	TX Registration Data from DFW Clean Cities (20...	
3	TX3	TX Registration Data from DFW Clean Cities (20...	
4	TX3	TX Registration Data from DFW Clean Cities (20...	

	Registration	Expiration Date	State Abbreviation	Geography	Vehicle Name	\
0		NaN	TX	ZIP Code	Tesla Model 3	
1		NaN	TX	ZIP Code	Tesla Model 3	
2		NaN	TX	ZIP Code	Tesla Model 3	
3		NaN	TX	ZIP Code	Tesla Model 3	
4		NaN	TX	ZIP Code	Tesla Model 3	

	Technology
0	BEV
1	BEV
2	BEV
3	BEV
4	BEV

```
[9]: # Keep the columns we need and check if there is any missing data
tx_ev = tx_ev[["State Abbreviation", "ZIP Code", "Vehicle Name"]]
tx_ev.isna().sum() # no null values
```

```
[9]: State Abbreviation    0
ZIP Code                  0
Vehicle Name              0
dtype: int64
```

```
[10]: # check the data types in this dataset
tx_ev.dtypes # the data type for ZIP Code column is object

# we need this column to be integers
tx_ev = tx_ev[tx_ev['ZIP Code'].apply(lambda x: x.isnumeric())]
tx_ev['ZIP Code'] = tx_ev['ZIP Code'].astype(int)
```

```
[11]: # groupby the dataset ny_ev by Zip code and Vehicle Name
# to get the counts of the vehicles by state
tx_vehicle_count = tx_ev.groupby('ZIP Code')['Vehicle Name']. \
    value_counts().rename('Vehicle_Counts').reset_index()

# create a new dataset counting the total number of evs in each zip code area
total_ev_tx = tx_vehicle_count.groupby('ZIP Code').sum(). \
    reset_index().rename(columns = {"ZIP Code": "Zip_Code",
    ↪ "Vehicle_Counts": "Total_EV_Counts"})

# create a new dataset showing the most popular models of ev in each zip code
↪ area
max_ev_tx = tx_vehicle_count.groupby('ZIP Code').max(). \
    reset_index().rename(columns = {"ZIP Code": "Zip_Code"})
```

Dataset #3

```
[12]: # import dataset #3
income = pd.read_csv("kaggle_income.csv")
```

```
[13]: # get an idea of how the data looks like
income.head()
```

```
[13]:
```

	id	State_Code	State_Name	State_ab	County	City	\
0	1011000	1	Alabama	AL	Mobile County	Chickasaw	
1	1011010	1	Alabama	AL	Barbour County	Louisville	
2	1011020	1	Alabama	AL	Shelby County	Columbiana	
3	1011030	1	Alabama	AL	Mobile County	Satsuma	
4	1011040	1	Alabama	AL	Mobile County	Dauphin Island	

	Place	Type	Primary	Zip_Code	Area_Code	ALand	AWater	\
0	Chickasaw city	City	place	36611	251	10894952	909156	
1	Clio city	City	place	36048	334	26070325	23254	
2	Columbiana city	City	place	35051	205	44835274	261034	
3	Creola city	City	place	36572	251	36878729	2374530	
4	Dauphin Island	Town	place	36528	251	16204185	413605152	

	Lat	Lon	Mean	Median	Stdev	sum_w
0	30.771450	-88.079697	38773	30506	33101	1638.260513
1	31.708516	-85.611039	37725	19528	43789	258.017685
2	33.191452	-86.615618	54606	31930	57348	926.031000
3	30.874343	-88.009442	63919	52814	47707	378.114619
4	30.250913	-88.171268	77948	67225	54270	282.320328

```
[14]: # drop columns we don't need
income = income.drop(columns = ['id', 'State_Code', 'State_Name', 'County',
    ↪ 'City', 'Place',
```

```

        'Type', 'Primary', 'Area_Code', 'ALand', 'AWater', 'Lat', 'Lon', 'Median',
        ↪ 'Stdev', 'sum_w'] )

```

```

# rename column to match the format
income = income.rename(columns = {'Mean': 'Mean_Income'})

```

```

[15]: # extract income data of NY from the whole dataset
ny_income = income[income['State_ab'] == 'NY']

```

```

[16]: # extract income data of TX from the whole dataset
tx_income = income[income['State_ab'] == 'TX']

```

Combining datasets for NY

```

[17]: # merging two datasets(total_ev_ny,ny_income) on the key 'Zip_Code'
ny_total_combined = pd.merge(total_ev_ny, ny_income, how = 'inner', on =
    ↪ ['Zip_Code'])

# there are 0s in the Mean_Income, we want to drop these rows since they will
    ↪ skew our data
ny_total_combined = ny_total_combined.loc[ny_total_combined['Mean_Income'] !=
    ↪ 0].reset_index(drop=True)

# making sure there are no na values
ny_total_combined.isna().sum()

# take a look at the cleaned dataset
ny_total_combined

```

```

[17]:
      Zip_Code  Total_EV_Counts  State_ab  Mean_Income
0         6390                63         NY         88039
1         6812                 9         NY         99464
2        10001             4074         NY         52434
3        10002             3828         NY        100415
4        10002             3828         NY         71282
...         ...                ...         ...         ...
2100       14904                552         NY         52066
2101       14905             1181         NY         63865
2102       14905             1181         NY         46504
2103       14905             1181         NY        100110
2104       16701                 2         NY         45992

```

[2105 rows x 4 columns]

```

[18]: # merging two datasets(max_ev_ny,ny_income) on the key 'Zip_Code'
ny_max_combined = pd.merge(max_ev_ny, ny_income, how = 'inner', on =
    ↪ ['Zip_Code'])

```



```

ny_max_combined = ny_max_combined.rename(columns = {'Vehicle Name':
↳ 'Vehicle_Name'})

# there are 0s in the Mean_Income, we want to drop these rows since they will
↳ skew our data
ny_max_combined = ny_max_combined.loc[ny_max_combined['Mean_Income'] != 0].
↳ reset_index(drop=True)

# making sure there are no na values
ny_max_combined.isna().sum()

# take a look at the cleaned dataset
ny_max_combined

```

```

[18]:      Zip_Code      Vehicle_Name  Vehicle_Counts  State_ab  Mean_Income
0         6390      Tesla Model S             33         NY      88039
1         6812  Toyota Prius Prime              2         NY      99464
2        10001  Volvo XC90 Plug In            646         NY      52434
3        10002  Volvo XC90 Plug In            803         NY     100415
4        10002  Volvo XC90 Plug In            803         NY      71282
...         ...         ...         ...         ...         ...
2100       14904  Toyota Prius Prime            161         NY      52066
2101       14905   Toyota RAV4 Prime            231         NY      63865
2102       14905   Toyota RAV4 Prime            231         NY      46504
2103       14905   Toyota RAV4 Prime            231         NY     100110
2104       16701  Toyota Prius Prime              1         NY      45992

```

[2105 rows x 5 columns]

Combining datasets for TX

```

[19]: # merging two datasets(total_ev_tx,tx_income) on the key 'Zip_Code'
tx_total_combined = pd.merge(total_ev_tx, tx_income, how = 'inner', on =
↳ ['Zip_Code'])

# there are 0s in the Mean_Income, we want to drop these rows since they will
↳ skew our data
tx_total_combined = tx_total_combined.loc[tx_total_combined['Mean_Income'] !=
↳ 0].reset_index(drop=True)

# making sure there are no na values
tx_total_combined.isna().sum()

# take a look at the cleaned dataset
tx_total_combined

```

```
[19]:
```

	Zip_Code	Total_EV_Counts	State_ab	Mean_Income
0	75001	1783	TX	81139
1	75002	6934	TX	159185
2	75002	6934	TX	135524
3	75002	6934	TX	88657
4	75002	6934	TX	54617
...	...	...	...	...
2139	79936	2069	TX	51509
2140	79938	2334	TX	51271
2141	79938	2334	TX	74291
2142	79938	2334	TX	59184
2143	79938	2334	TX	49833

[2144 rows x 4 columns]

```
[20]: # merging two datasets(max_ev_tx,tx_income) on the key 'Zip_Code'
tx_max_combined = pd.merge(max_ev_tx, tx_income, how = 'inner', on =
    ↳['Zip_Code'])
tx_max_combined = tx_max_combined.rename(columns = {'Vehicle Name':
    ↳'Vehicle_Name'})

# there are 0s in the Mean_Income, we want to drop these rows since they will
    ↳skew our data
tx_max_combined = tx_max_combined.loc[tx_max_combined['Mean_Income'] != 0].
    ↳reset_index(drop=True)

# making sure there are no na values
tx_max_combined.isna().sum()

# take a look at the cleaned dataset
tx_max_combined
```

```
[20]:
```

	Zip_Code	Vehicle_Name	Vehicle_Counts	State_ab	Mean_Income
0	75001	Volvo XC60 Plug In	676	TX	81139
1	75002	Volvo XC90 Plug In	1929	TX	159185
2	75002	Volvo XC90 Plug In	1929	TX	135524
3	75002	Volvo XC90 Plug In	1929	TX	88657
4	75002	Volvo XC90 Plug In	1929	TX	54617
...	...	...	...	...	...
2139	79936	Volvo XC90 Plug In	492	TX	51509
2140	79938	Volvo XC90 Plug In	574	TX	51271
2141	79938	Volvo XC90 Plug In	574	TX	74291
2142	79938	Volvo XC90 Plug In	574	TX	59184
2143	79938	Volvo XC90 Plug In	574	TX	49833

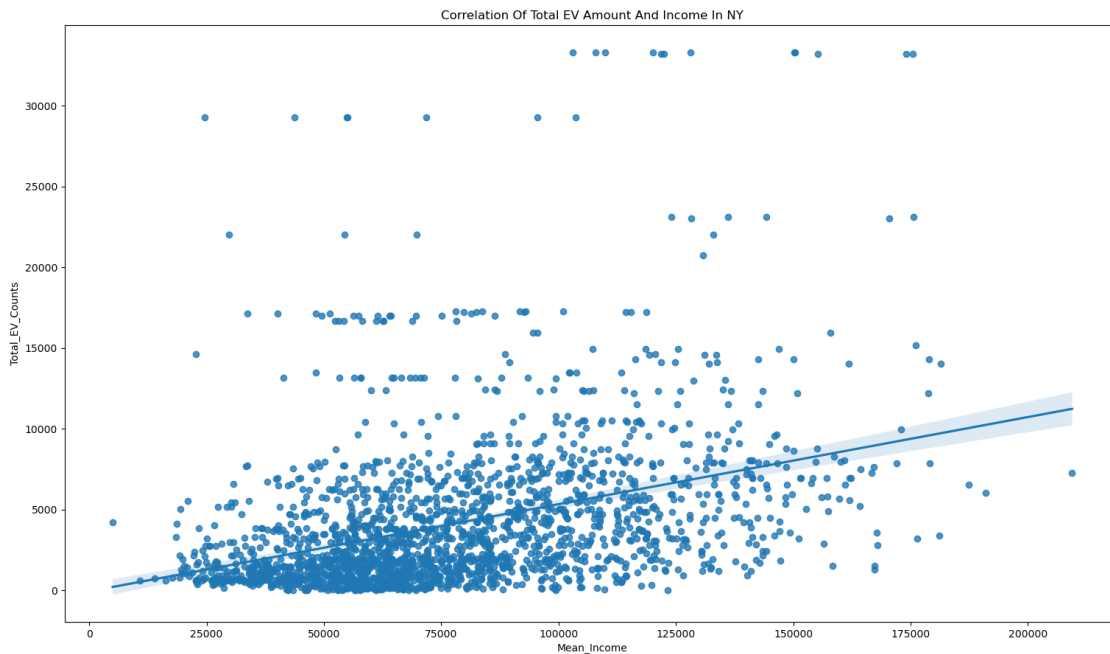
[2144 rows x 5 columns]

9 Data Analysis & Results

Correlation of total ev amount and income in NY

```
[21]: # Generate a scatter plot to see the correlation
# between the total ev amount and house income in the state of New York
fig1, ax = plt.subplots()
fig1.set_size_inches(18.5, 10.5)
sns.regplot(data = ny_total_combined, x = "Mean_Income", y = "Total_EV_Counts")\
    .set(title='Correlation Of Total EV Amount And Income In NY')
```

```
[21]: [Text(0.5, 1.0, 'Correlation Of Total EV Amount And Income In NY')]
```



This scatterplot graph shows the correlation between the Total EV Amount and Income In NY state. The x-axis is the mean income by zipcode area, the y-axis is the total electric vehicle counts by zipcode area. From the graph we can see that there is a positive correlation between these two values, which means that the higher income zipcode areas are more likely to have more electric vehicles.

```
[43]: #'Correlation Of Total EV Amount And Income In NY' analysis
model = sm.formula.ols(formula='Total_EV_Counts ~ Mean_Income',
    data=ny_total_combined)
res_1 = model.fit()
print(res_1.summary())
```

OLS Regression Results

```
=====
Dep. Variable:          Total_EV_Counts    R-squared:          0.141
```

```

Model:                OLS      Adj. R-squared:      0.141
Method:               Least Squares    F-statistic:      346.6
Date:                 Thu, 01 Dec 2022    Prob (F-statistic):    9.94e-72
Time:                 19:34:05    Log-Likelihood:      -20596.
No. Observations:     2105    AIC:                4.120e+04
Df Residuals:         2103    BIC:                4.121e+04
Df Model:              1
Covariance Type:      nonrobust

```

```

=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
Intercept    -66.9951     244.594     -0.274      0.784    -546.667     412.677
Mean_Income    0.0540       0.003     18.617      0.000      0.048      0.060
=====
Omnibus:                 1368.079    Durbin-Watson:           0.537
Prob(Omnibus):            0.000    Jarque-Bera (JB):        17257.536
Skew:                     2.930    Prob(JB):                 0.00
Kurtosis:                 15.744    Cond. No.                 2.20e+05
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.2e+05. This might indicate that there are strong multicollinearity or other numerical problems.

From the regression results, we can see that the R-squared value(coefficient of determination) is 0.141 which indicates that the model explains 14.1% of the data. This value also tells us that the correlation between the Total EV Amount and Income In NY state is not super strong. - R-squared is always between 0 and 1: 0 indicates that the model does not fit the dataset, 1 indicates that the model fits the data perfectly.

Correlation of total ev amount and income in TX

```

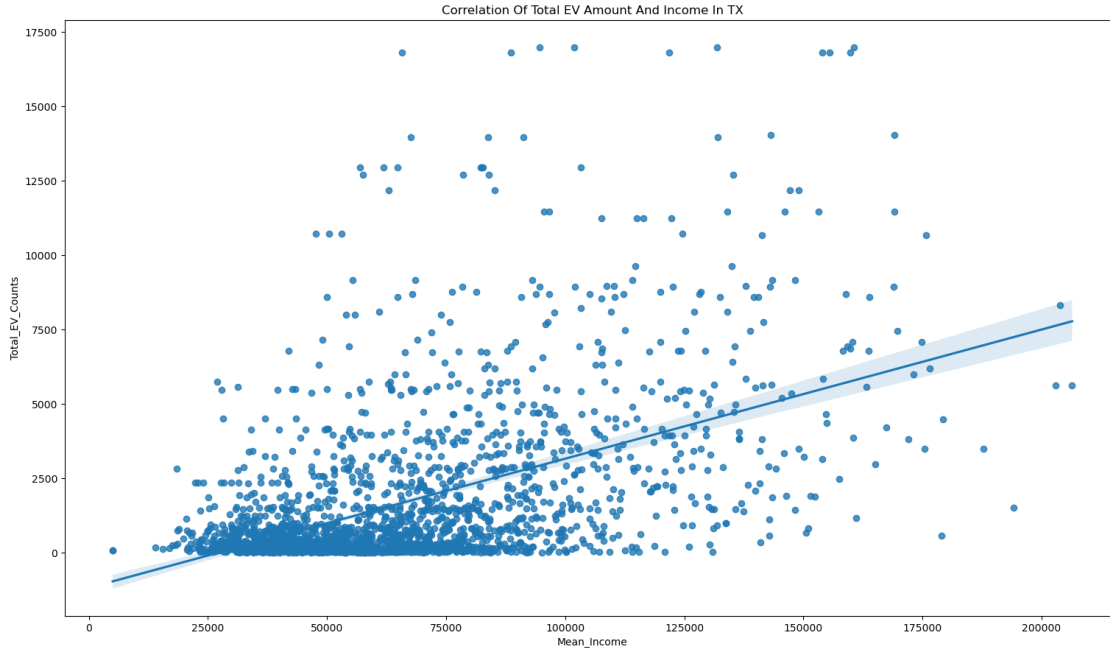
[22]: # Generate a scatter plot to see the correlation
      # between the total ev amount and house income in the state of Texas
      fig2, ax = plt.subplots()
      fig2.set_size_inches(18.5, 10.5)
      sns.regplot(data = tx_total_combined, x = "Mean_Income", y = "Total_EV_Counts")\
        .set(title='Correlation Of Total EV Amount And Income In TX')

```

```

[22]: [Text(0.5, 1.0, 'Correlation Of Total EV Amount And Income In TX')]

```



This scatterplot graph shows the correlation between the Total EV Amount and Income In TX state. The x-axis is the mean income by zipcode area, the y-axis is the total electric vehicle counts by zipcode area. From the graph we can see that there is a positive correlation between these two values, which means that the higher income zipcode areas are more likely to have more electric vehicles. Moreover, this graph looks very similar to the graph for NY state.

```
[41]: #'Correlation Of Total EV Amount And Income In TX' analysis
model = sm.formula.ols(formula='Mean_Income ~ Total_EV_Counts',
    data=tx_total_combined)
res_2 = model.fit()
print(res_2.summary())
```

OLS Regression Results

```
=====
Dep. Variable:          Mean_Income    R-squared:                0.258
Model:                  OLS            Adj. R-squared:          0.258
Method:                 Least Squares   F-statistic:              745.2
Date:                   Thu, 01 Dec 2022 Prob (F-statistic):       4.63e-141
Time:                   19:27:00        Log-Likelihood:           -24879.
No. Observations:       2144            AIC:                     4.976e+04
Df Residuals:           2142            BIC:                     4.977e+04
Df Model:                1
Covariance Type:        nonrobust
=====
```

```
=====
coef    std err          t      P>|t|      [0.025
0.975]
```

```

-----
---
Intercept          5.662e+04    683.812    82.795    0.000    5.53e+04
5.8e+04
Total_EV_Counts    5.9434      0.218    27.299    0.000    5.516
6.370
=====
Omnibus:                260.925    Durbin-Watson:                1.518
Prob(Omnibus):          0.000    Jarque-Bera (JB):            450.075
Skew:                   0.811    Prob(JB):                    1.85e-98
Kurtosis:               4.552    Cond. No.                    3.75e+03
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.75e+03. This might indicate that there are strong multicollinearity or other numerical problems.

From the regression results, we can see that the R-squared value(coefficient of determination) is 0.258 which indicates that the model explains 25.8% of the data. This value also tells us that the correlation between the Total EV Amount and Income In TX state is not super strong. However, this correlation is much stronger than it is in NY. - R-squared is always between 0 and 1: 0 indicates that the model does not fit the dataset, 1 indicates that the model fits the data perfectly.

Total number of EVs in NY distribution graph

```

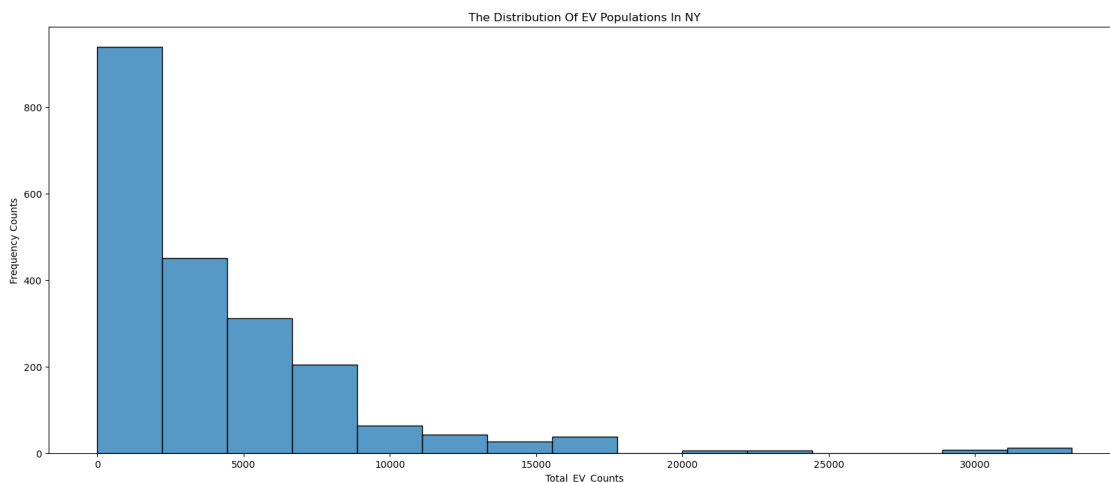
[47]: # use a histogram to show the distribution of total ev amounts
# by zip codes
f1,ax = plt.subplots(figsize =(20, 8))
sns.histplot(data = ny_total_combined, x = 'Total_EV_Counts', bins = 15)\
        .set(title='The Distribution Of EV Populations In NY')
plt.ylabel('Frequency Counts')

```

```

[47]: Text(0, 0.5, 'Frequency Counts')

```

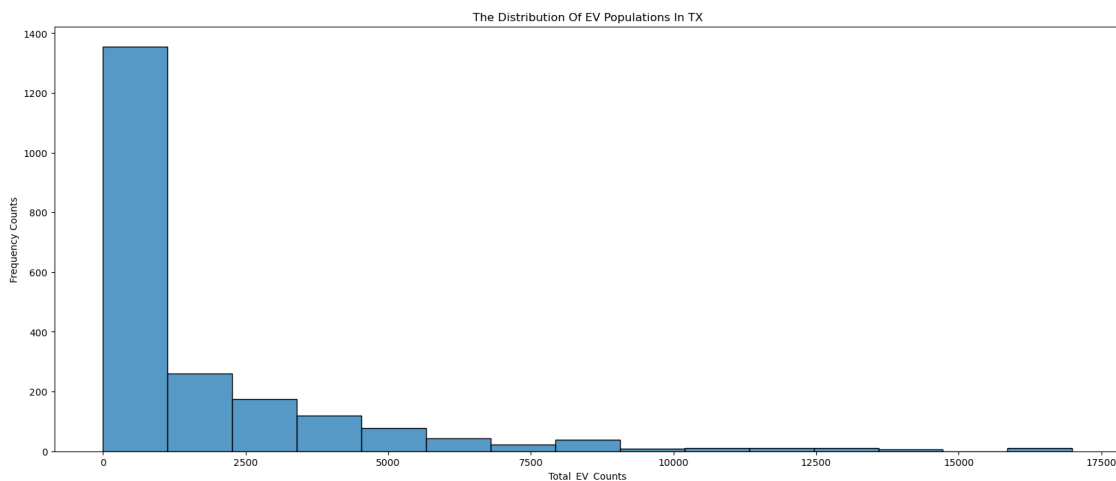


This graph shows the distribution of EV populations in New York state. From this graph we can see that most zipcode areas in New York State have a total electric vehicle population from 0 to 10000

Total number of EVs in TX distribution graph

```
[46]: # use a histogram to show the distribution of total ev amounts
# by zip codes
f2,ax = plt.subplots(figsize =(20, 8))
sns.histplot(data = tx_total_combined, x = 'Total_EV_Counts', bins = 15)\
        .set(title='The Distribution Of EV Populations In TX')
plt.ylabel('Frequency Counts')
```

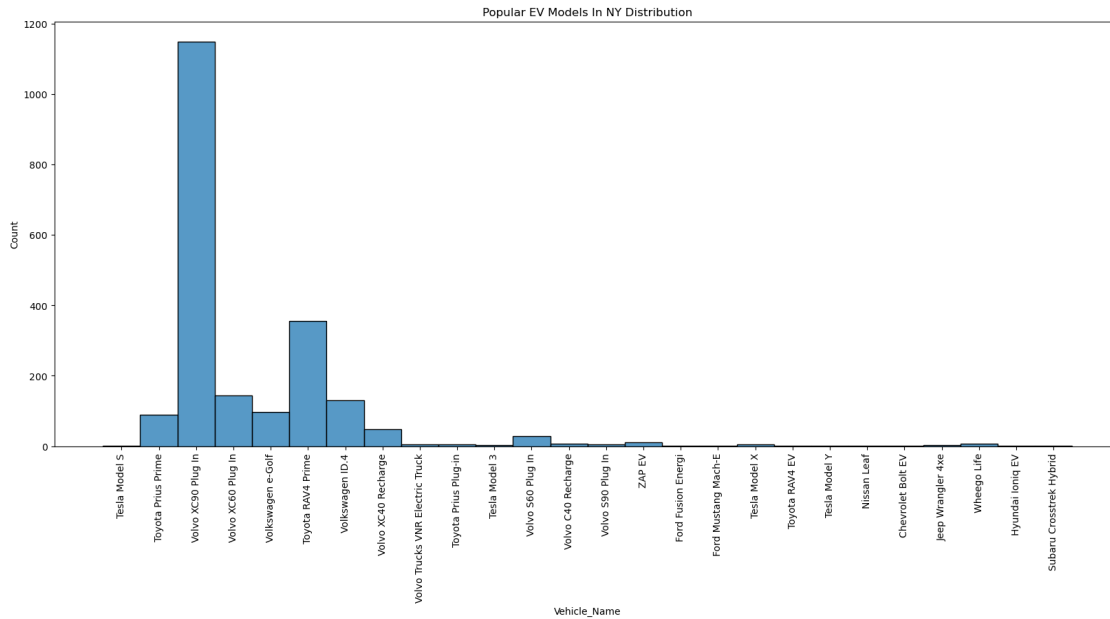
```
[46]: Text(0, 0.5, 'Frequency Counts')
```



This graph shows a distribution of EV populations in Texas. From this graph we can see that most zipcode areas in Texas have a total electric vehicle population from 0 to 2500, this is an interesting observation because this number is significantly less than the distribution of EV populations in New York state

Popular EV models in NY distribution graph

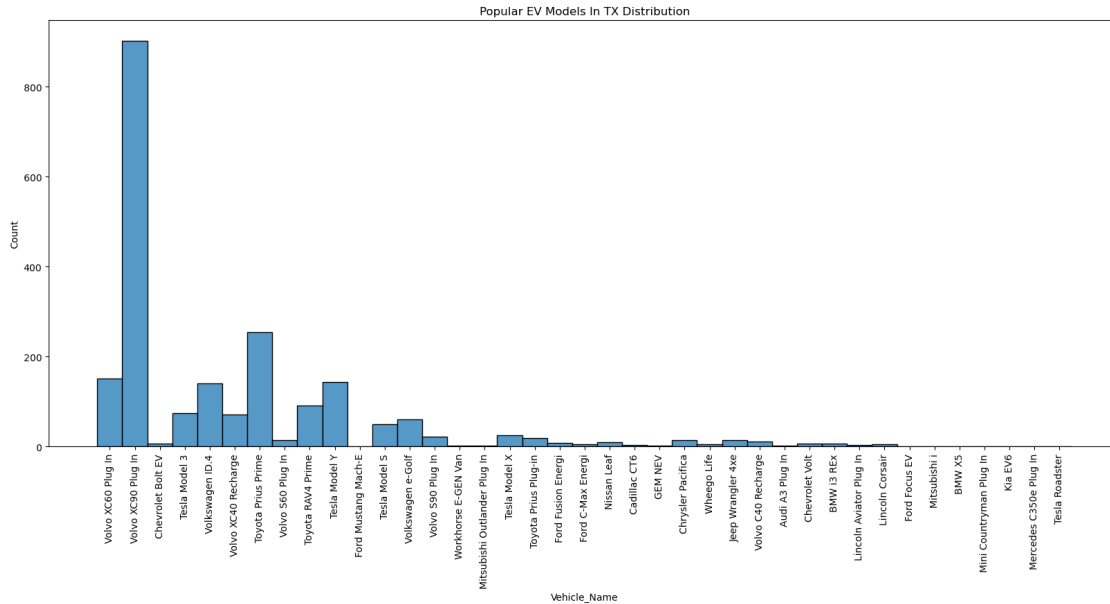
```
[25]: # use a histogram to show the distribution of
# the popular models in NY
f3,ax = plt.subplots(figsize =(20, 8))
sns.histplot(ny_max_combined['Vehicle_Name']).set(title='Popular EV Models In_
↳NY Distribution')
ax.tick_params(axis='x', rotation=90)
```



This graph shows the most popular models in New York State. It appears that Volvo XC90 Plug In is the most popular model in the state of New York. There are 26 most popular models in the state of New York

Popular EV models in TX distribution graph

```
[26]: # use a histogram to show the distribution of
# the popular models in NY
f4,ax = plt.subplots(figsize=(20, 8))
sns.histplot(tx_max_combined['Vehicle_Name']).set(title='Popular EV Models In_
↳TX Distribution')
ax.tick_params(axis='x', rotation=90)
```

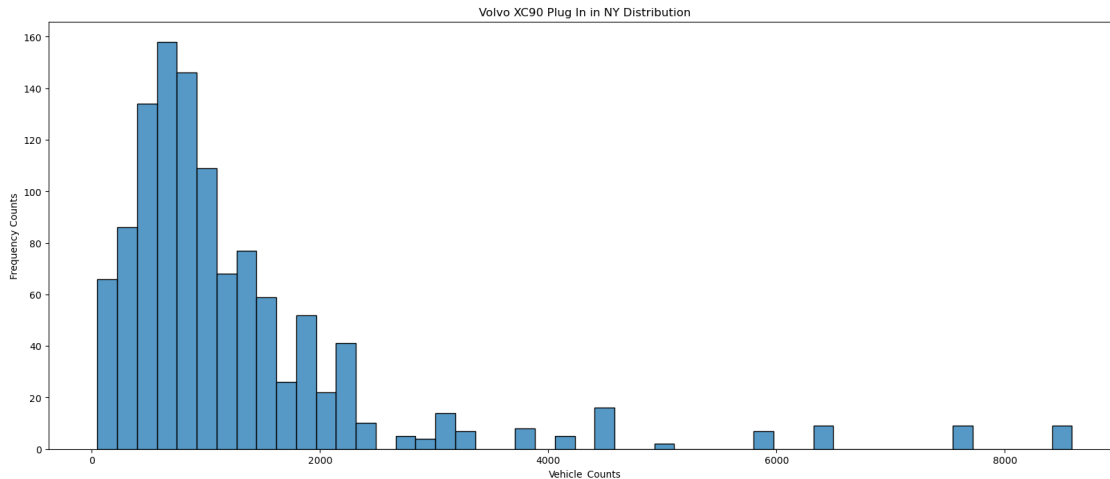



This graph shows the most popular models in Texas. It appears that Volvo XC90 Plug In is also the most popular model in the state of Texas. There are 39 most popular models in the state of Texas, which is 13 models more than New York's most popular models. This shows that there is more variety in the most popular electric vehicle models in the state of Texas

Volvo XC90 Plug In in NY distribution graph

```
[44]: # from the graphs above we see that 'Volvo XC90 Plug In' is the most popular_
      ↪model
      # among all the popular EV models in NY
      # use a histogram to see the distribution of this model in NY
      xc90_ny = ny_max_combined.loc[ny_max_combined['Vehicle_Name'] == 'Volvo XC90_
      ↪Plug In']
      f5,ax = plt.subplots(figsize=(20, 8))
      sns.histplot(xc90_ny['Vehicle_Counts']).set(title='Volvo XC90 Plug In in NY_
      ↪Distribution')
      plt.ylabel('Frequency Counts')
```

```
[44]: Text(0, 0.5, 'Frequency Counts')
```

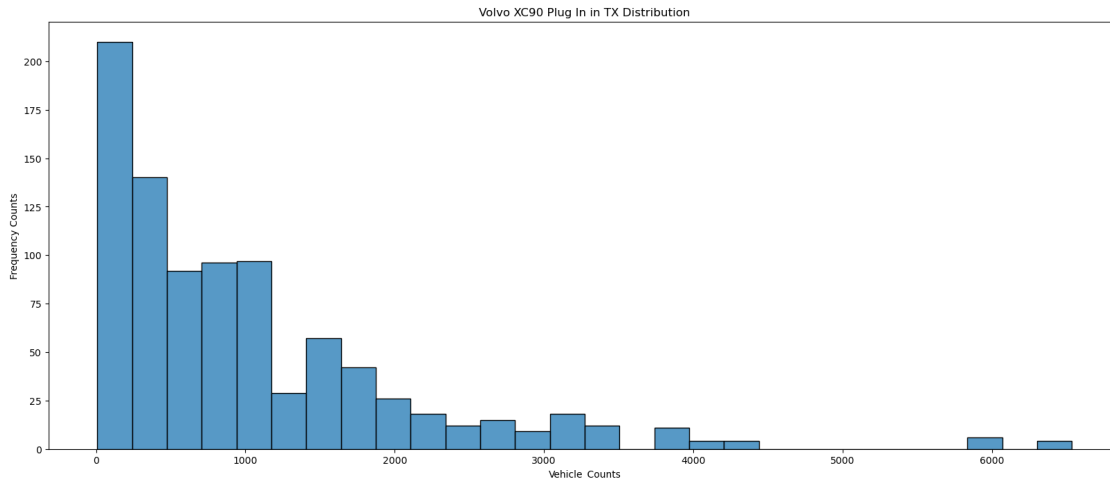


This graph shows the distribution of Volvo XC90 Plug In populations in New York state. From this graph we can see that most zipcode areas in New York State have a total Volvo XC90 Plug In population from 0 to 2000

Volvo XC90 Plug In in TX distribution graph

```
[45]: # from the graphs above we see that 'Volvo XC90 Plug In' is the most popular_
      ↪model
      # among all the popular EV models in TX
      # use a histogram to see the distribution of this model in TX
xc90_tx = tx_max_combined.loc[tx_max_combined['Vehicle_Name'] == 'Volvo XC90_
      ↪Plug In']
f5,ax = plt.subplots(figsize =(20, 8))
sns.histplot(xc90_tx['Vehicle_Counts']).set(title='Volvo XC90 Plug In in TX_
      ↪Distribution')
plt.ylabel('Frequency Counts')
```

```
[45]: Text(0, 0.5, 'Frequency Counts')
```

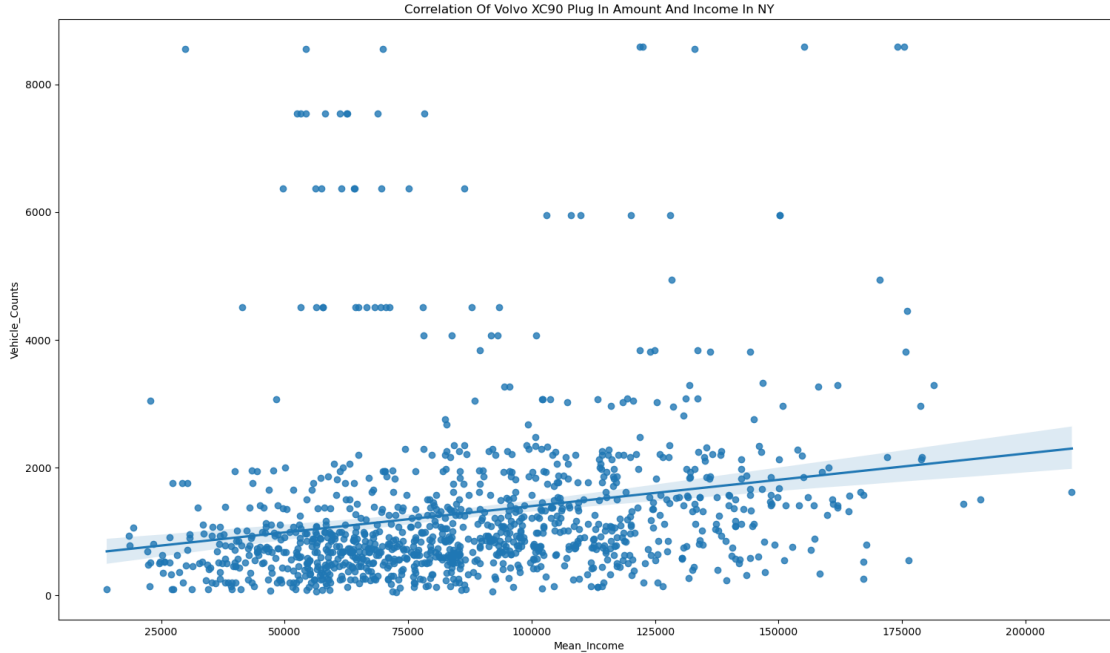


This graph shows the distribution of Volvo XC90 Plug In populations in the state of Texas. From this graph we can see that most zipcode areas in New York State have a total Volvo XC90 Plug In population from 0 to 2000

Correlation Of Volvo XC90 Plug In Amount And Income In NY graph

```
[29]: # Generate a scatter plot to see the correlation
# between the Volvo XC90 Plug In amount and house income in the state of New
      ↪York
fig3, ax = plt.subplots()
fig3.set_size_inches(18.5, 10.5)
sns.regplot(data = xc90_ny, x = "Mean_Income", y = "Vehicle_Counts")\
          .set(title='Correlation Of Volvo XC90 Plug In Amount And Income
      ↪In NY')
```

```
[29]: [Text(0.5, 1.0, 'Correlation Of Volvo XC90 Plug In Amount And Income In NY')]
```



This scatterplot graph shows the correlation between the Volvo XC90 Plug In Amount and Income In NY state. The x-axis is the mean income by zipcode area, the y-axis is the the Volvo XC90 Plug In counts by zipcode area. From the graph we can see that there is a positive correlation between these two values, which means that the higher income zipcode areas are more likely to have the most popular model.

```
[36]: # 'Correlation Of Volvo XC90 Plug In Amount And Income In NY' analysis
model = sm.formula.ols(formula='Vehicle_Counts ~ Mean_Income', data=xc90_ny)
res_3 = model.fit()
print(res_3.summary())
```

OLS Regression Results

```
=====
Dep. Variable:      Vehicle_Counts      R-squared:                0.043
Model:              OLS                 Adj. R-squared:           0.042
Method:             Least Squares       F-statistic:             51.61
Date:               Thu, 01 Dec 2022    Prob (F-statistic):      1.21e-12
Time:               19:04:29            Log-Likelihood:          -9871.4
No. Observations:   1149                AIC:                     1.975e+04
Df Residuals:       1147                BIC:                     1.976e+04
Df Model:           1
Covariance Type:    nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	573.3955	107.039	5.357	0.000	363.382	783.409
Mean_Income	0.0082	0.001	7.184	0.000	0.006	0.010

```
=====
Omnibus:                808.135    Durbin-Watson:                0.375
Prob(Omnibus):           0.000    Jarque-Bera (JB):           9409.928
Skew:                    3.243    Prob(JB):                   0.00
Kurtosis:                15.429    Cond. No.                   2.60e+05
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

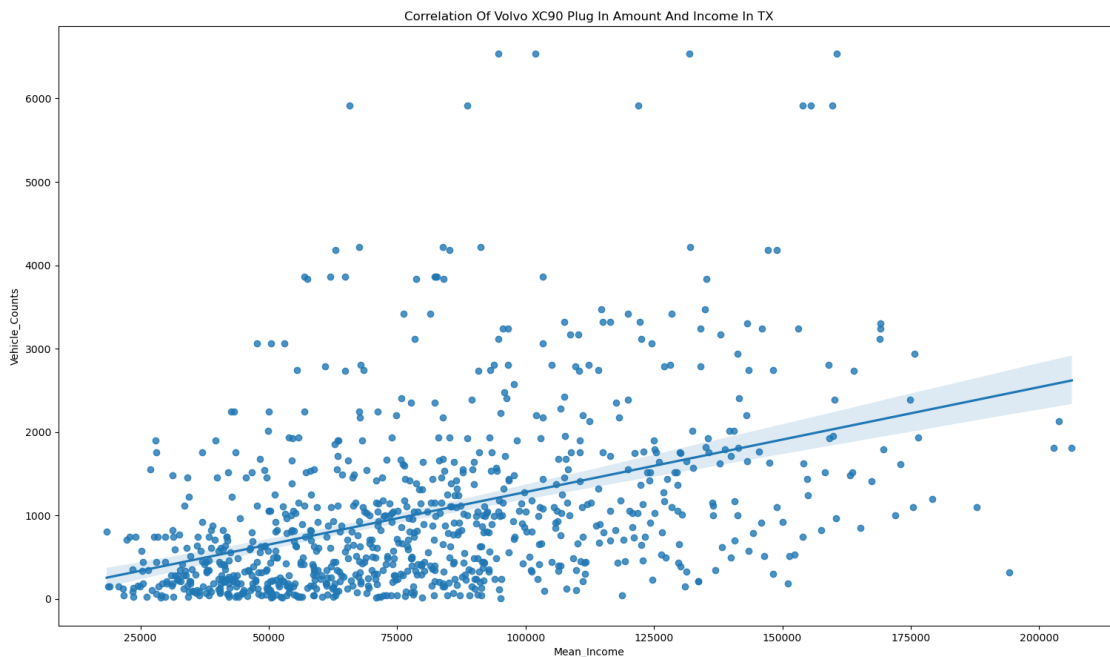
[2] The condition number is large, 2.6e+05. This might indicate that there are strong multicollinearity or other numerical problems.

From the regression results, we can see that the R-squared value(coefficient of determination) is 0.043 which indicates that the model explains 4.3% of the data. This value also tells us that the correlation between the Volvo XC90 Plug In counts and Income In NY state is very weak. - R-squared is always between 0 and 1: 0 indicates that the model does not fit the dataset, 1 indicates that the model fits the data perfectly.

Correlation Of Volvo XC90 Plug In Amount And Income In TX graph

```
[30]: # Generate a scatter plot to see the correlation
# between the Volvo XC90 Plug In amount and house income in the state of Texas
fig4, ax = plt.subplots()
fig4.set_size_inches(18.5, 10.5)
sns.regplot(data = xc90_tx, x = "Mean_Income", y = "Vehicle_Counts")\
    .set(title='Correlation Of Volvo XC90 Plug In Amount And Income_
↪In TX')
```

```
[30]: [Text(0.5, 1.0, 'Correlation Of Volvo XC90 Plug In Amount And Income In TX')]
```



This scatterplot graph shows the correlation between the Volvo XC90 Plug In Amount and Income In Texas. The x-axis is the mean income by zipcode area, the y-axis is the the Volvo XC90 Plug In counts by zipcode area. From the graph we can see that there is a positive correlation between these two values, which means that the higher income zipcode areas are more likely to have the most popular model.

```
[37]: # 'Correlation Of Volvo XC90 Plug In Amount And Income In TX' analysis
model = sm.formula.ols(formula='Vehicle_Counts ~ Mean_Income', data=xc90_tx)
res_4 = model.fit()
print(res_4.summary())
```

```

                                OLS Regression Results
=====
Dep. Variable:          Vehicle_Counts      R-squared:                0.180
Model:                  OLS                 Adj. R-squared:            0.179
Method:                 Least Squares       F-statistic:              196.9
Date:                  Thu, 01 Dec 2022     Prob (F-statistic):       1.32e-40
Time:                  19:04:54             Log-Likelihood:           -7476.0
No. Observations:      902                 AIC:                     1.496e+04
Df Residuals:          900                 BIC:                     1.497e+04
Df Model:               1
Covariance Type:       nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	23.3817	78.540	0.298	0.766	-130.762	177.525
Mean_Income	0.0126	0.001	14.034	0.000	0.011	0.014

```

=====
Omnibus:                 398.271    Durbin-Watson:              0.631
Prob(Omnibus):            0.000    Jarque-Bera (JB):           1973.403
Skew:                     2.018    Prob(JB):                   0.00
Kurtosis:                 9.018    Cond. No.:                  2.15e+05
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.15e+05. This might indicate that there are strong multicollinearity or other numerical problems.

From the regression results, we can see that the R-squared value(coefficient of determination) is 0.180 which indicates that the model explains 18% of the data. This value also tells us that the correlation between the Volvo XC90 Plug In counts and Income In Texas is weak but this is a lot higher than it is in the state of New York. From this observation we can see that the correlation between the Volvo XC90 Plug In counts and Income in Texas is much stronger than the correlation between the Volvo XC90 Plug In counts and Income in the state of New York. -

R-squared is always between 0 and 1: 0 indicates that the model does not fit the dataset, 1 indicates that the model fits the data perfectly.

Description

- 1) What distributions do your variables take?
 - When comparing the total EV amount and income of NY and TX we can see that most of our data lies between a mean income of \$25,000 and \$100,000. If we take a glance at the distribution of EV populations in NY and TX as they relate to zip code, we can see that the majority of EV registrations for the majority of the zip codes lie between 0-5000 for NY and 0-2500 for TX. When comparing brand popularity, we can see that the majority of the data falls into the Volvo XC90 plug-in category and appears to be the most popular car between the two states. Looking at the Volvo XC90 plug in distributions in each state, we can see that most of the data lies between 0-2000 for NY and 0-1000 for TX.
- 2) Are there any outliers?
 - As for outliers, we eliminated any outliers that would significantly skew our data during the cleaning stage.
- 3) Relationship between variables?
 - First, we want to see how total EV amount relates to income in the states of NY and TX. Going further, we can examine how the distribution of EV populations in NY and TX relate to zip code. Then, we want to see which EV brand is most popular in each state. From here, we can then see how the most popular brand (Volvo XC90 plug-in) is distributed in the states of NY and TX. Additionally, we generated some scatterplots to view the correlation between Volvo XC90 plug-ins and house income in the states of NY and TX.

10 Ethics & Privacy

- As data scientists, we must be cognizant and respectful of other people's information. In doing this project, our priority was protecting participants' private information. Our datasets are coming from public sources, it was obtained through Atlas EV Hub, Atlas EV Hub works with states to make data on electric vehicle registrations publicly accessible. Which means providing clear and transparent information about how the data will be collected and used and giving individuals the opportunity to opt out if they do not want their data to be included. And making this information available to the public. This can help to build trust and accountability and can also help individuals to understand how their data is being used. Moreover, developing and implementing strong security measures to prevent unauthorized access to datasets. We are only using it for the purpose of the study, which means only people who have authorized access can view our work; it's not accessible to outsiders. Clearly defining the purpose of the dataset and ensuring that it is being used for legitimate and ethical purposes. Last, Data is only included from states that gave consent and choose to participate in the Atlas EV Hub's initiative. All the datasets that we used do not contain individuals' personal information; To eliminate the concern of sharing anything personal only the first eight to eleven digits of the VIN number is included in the data to avoid any ethical and privacy violations. The datasets only provide data based on states and zip code areas. It

ensures that personal information cannot be easily linked back to specific individuals. That being said the hard work was already done for us and the data is in ethical form.

11 Conclusion & Discussion

- In conclusion, once our data was cleaned and analyzed, we were able to utilize our visualizations and inferential statistics to reject our null hypothesis and accept our alternative hypothesis, which in turn supported our general hypothesis. With our statistically significant results, we accept that a higher home income influences a higher population of electric vehicles in both the state of New York and Texas, while a lower home income influences a lower population of electric vehicles in the state of New York and Texas. We reject that there is no difference between the correlation of the number of Electric and Hybrid vehicles and the mean house income in the state of New York and Texas. As a result, we can say that there is a correlation between the number of Electric and Hybrid vehicles registered and house income in both the state of New York and Texas.
- The above findings were determined from the regression results in both states. In terms of New York, we were able to see that the R-squared value was 0.043, which indicated that the correlation between the Volvo XC90 Plug-In population and housing income in the state of New York is very weak. Similarly, we found that the R-squared value in Texas state was 0.180, which displayed that the correlation between the Volvo XC90 Plug-In population and housing income in the state of Texas was weak but much stronger than that of the state of New York.
- However, we must recognize that since these R-squared values are rather low for both Texas and New York, only a small amount of our data actually fits our model. This means that there is a low variance in the relationship between our independent variable (house income) and our dependent variable (vehicle population). Believe it or not, this isn't necessarily a bad thing. Before the cleaning stage, our data was messy and consisted of millions of observations. In other words, our data was rather noisy and highly variable, yet we were still able to prove a significant trend between the relationship of variables which was the correlation exhibited in the scatterplots.
- Moreover, we were able to see a positive correlation in each of the scatterplots that aligned with our overall findings. Furthermore, from the histograms above, it appears that the most popular model of the electric vehicle seems to be the Volvo XC-90 Plug in both states of interest. In terms of distribution, we can see that the majority of the population of Volvo XC-90 Plug-in lies between 0 and 2000 vehicles in the state of New York and 0 and 1000 vehicles in the state of Texas. We also observed that Texas has a lower vehicle population as opposed to New York. However, there is more variation of popular electric and hybrid vehicle models in Texas as opposed to New York.

12 Limitations:

- There were several limitations to our research question. First, we only selected two states, and that kind of restricted us from getting what other states would be in terms of the house income and number of EVs, would be the same popular EVs or not. More state data would give us a closer result. Second, our initial plan was also to find the gas price as one variable

to see if there is a relation. However, we couldn't find the data we needed. So we only have income as the comparable variable to variable Evs in two states, but there could be other facts that are also involved in the number of Evs or the most popular brands. Third, our data is not based on a timeline; it's about the area, so that could also potentially skip some information such as when these results represent, is it recent years or 20? 30 years ago? Can the results be used to help us understand the market now? Last, the methods we used for the analysis are not broad enough to allow us to see the results from different angles. Because of the data and research question we have, we mainly used scatter plots, barplot, and regression to see the relations. It's certain that there is more that can be done to make this research more complete and useful for others as a reference.

13 Team Contributions

- Michael Schaar: Date and topic research, data analysis, and cleaning. Writing graph descriptions, research questions, and conclusions. Participating in team meetings for project research.
- Saman Zarei: Writing dataset descriptions, dataset research, and Ethics & Privacy. Data clearing, coding graphs.
- Shan He: Data analysis, data cleaning, dataset research, graph descriptions, Participating in team meetings for project research.
- Bo li: Researching the right datasets for the projects, and participating in team meetings for project research. Writing background, research questions, limitations, analysis, and clear data.